12 Change of Variable

A/Prof. Vera Roshchina

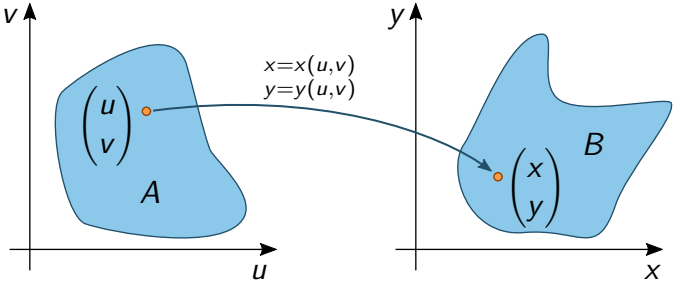
School of Mathematics and Statistics
UNSW Sydney

MATH2111 Several Variable Calculus, T1 2025

Intuition behind the change of variables

Suppose that $B = \mathbf{F}(A)$, where $A \subset \mathbb{R}^2$, $\mathbf{F} : A \to \mathbb{R}^2$, and

$$\mathbf{F}(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}.$$

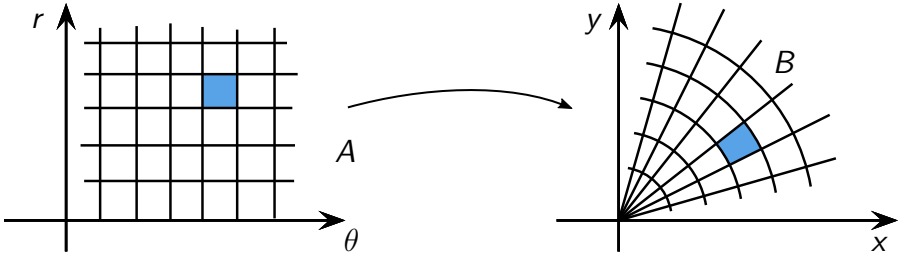


We compose $f : B \to \mathbb{R}$ with the change of variables function $\mathbf{F}$ to get a new function $g(u, v) = f(\mathbf{F}(u, v)) = f(x(u, v), y(u, v))$ defined on A.

Example: polar coordinates

We may want to change from the Cartesian coordinates to another set of coordinates in which the region has a more convenient expression. For instance, we can use the polar coordinates. Let

$$x = r\cos\theta, \quad y = r\sin\theta \quad (r \geq 0,\ \theta \in [0, 2\pi]).$$



To integrate using the new variables we need to adjust our 'elementary volume' appropriately.

Change of variables

The Jacobian determinant $\det J_{\mathbf{F}}$ measures how much $\mathbf{F}$ is expanding or contracting the volume around each point.

Theorem 12.1

Let $A \subset \mathbb{R}^n$ be an open set and let $\mathbf{F} : A \to \mathbb{R}^n$ be a one-to-one (injective) continuously differentiable function such that $\det J_{\mathbf{F}}(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in A$. If $f : \mathbf{F}(A) \to \mathbb{R}^n$ is an integrable function, then

$$\int_{\mathbf{F}(A)} f = \int_A f \circ \mathbf{F} \, |\det J_{\mathbf{F}}|.$$

Note: $|\det J_{\mathbf{F}}|$ is the absolute value of the Jacobian determinant.

Theorem 12.2 (Sard)

Let $\mathbf{F} : A \to \mathbb{R}^n$ be continuously differentiable, where $A \subset \mathbb{R}^n$ is open. Then the image of the points of A on which the determinant of $J_{\mathbf{F}}$ is zero has measure 0.

The Jacobian of the polar change of variables

For the polar coordinates, the coordinate transform $\mathbf{F}$ is

$$\mathbf{F}\begin{pmatrix} r \\ \theta \end{pmatrix} = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$$

We can compute the Jacobian determinant:

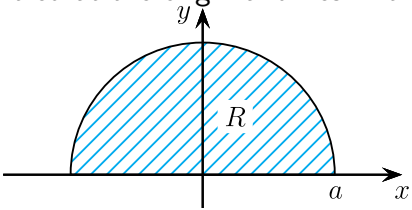
$$\begin{aligned} \det J_{\mathbf{F}}(r,\theta) &= \det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} \\ &= r\cos^2\theta + r\sin^2\theta = r. \end{aligned}$$

Note: even though the Jacobian is zero when $r = 0$, we can still apply Theorem 12.1, since we can remove this point from the region of integration without impacting the value of the integral.

Example 12.3

Find the centroid of a semi-circle of radius a using polar coordinates.

Assume the semi-circle is centred at the origin and lies in the upper half plane:



The area is $\frac{1}{2}\pi a^2$, and the centroid must lie on the y-axis, that is the x-coordinate of the centroid is $\bar{x} = 0$.

We only need to find the $y-$moment, $\iint_R y\, dA$.

We will use polars to calculate this integral.

Example 12.4

Find the volume of the solid under the surface $z = e^{x^2+y^2}$ above the polar rectangle given by

$$1 \leq r \leq 3, \quad 0 \leq \theta \leq \frac{1}{4}\pi.$$

Just as in the case of Cartesian coordinates, we can integrate over more complicated regions defined by polar coordinates by using variable integration limits.

Example 12.5

Sketch the four-leaved rose, given in polars by $r = 2\sin 2\theta$ and calculate the area of one of its leaves.

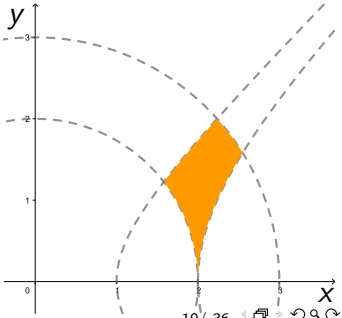
Example 12.6

Evaluate

$$\iint_B xy\,dx\,dy,$$

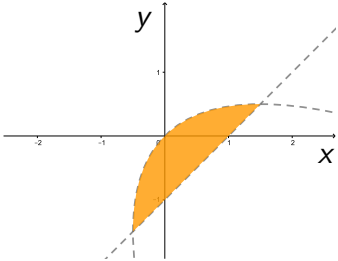
where B is the first-quadrant region bounded by the curves

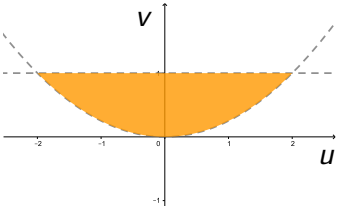
$$x^2 + y^2 = 4,\ x^2 + y^2 = 9,\ x^2 - y^2 = 1,\ x^2 - y^2 = 4.$$



Example 12.7

By making the substitutions $u = x + y$ and $v = x - y$ determine the area of the region bounded by the curves $(x + y)^2 = 4(x - y)$ and $x - y = 1$.





Example 12.8

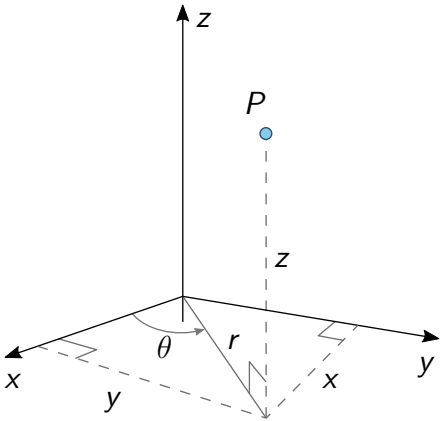
Find the volume of the region bounded by the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

where $a, b, c > 0$.

Cylindrical coordinates

When the region of integration D has an axis of symmetry, it may be convenient to do the integration using cylindrical coordinates.

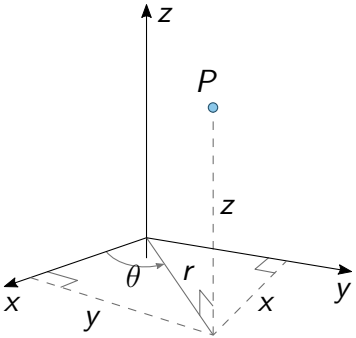


Change from the Cartesian to cylindrical Coordinates

The coordinate change from Cartesian (x, y, z) to cylindrical (r, θ, z) is given by

$$x = r\cos\theta,$$
$$y = r\sin\theta,$$
$$z = z.$$

Note that $r \geq 0$, $\theta \in [0, 2\pi]$.

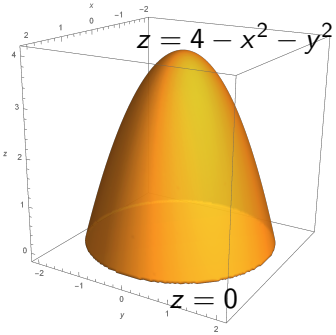


Example 12.9

Express the region D defined by the inequalities

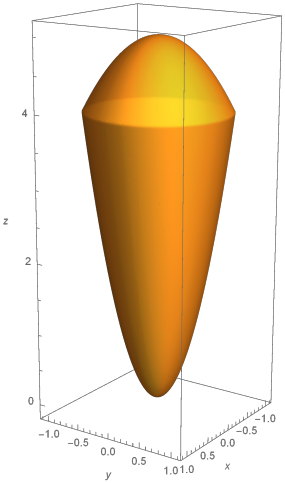
$$-2 \le x \le 2,$$
$$-\sqrt{4-x^2} \le y \le \sqrt{4-x^2},$$
$$0 \le z \le 4 - x^2 - y^2,$$

in cylindrical coordinates.

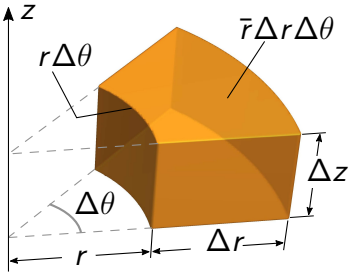


Example 12.10

Sketch the region enclosed between the two paraboloids $z = 4x^2 + 4y^2$ and $z = 5 - x^2 - y^2$, and express this region in cylindrical coordinates.



The Jacobian calculation for cylindrical coordinates



The coordinate transform is

$$\mathbf{F}\begin{pmatrix} r \\ \theta \\ z \end{pmatrix} = \begin{pmatrix} r\cos\theta \\ r\sin\theta \\ z \end{pmatrix}, \text{ then}$$

$$|J_{\mathbf{F}}(r,\theta,z)| = \left|\det\begin{pmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}\right| = |r| = r.$$

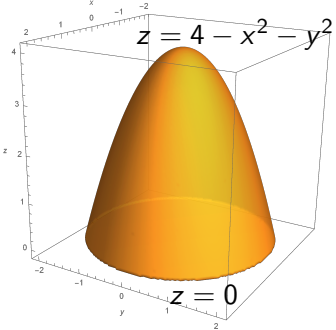
Example 12.11 (cf. Example 12.9)

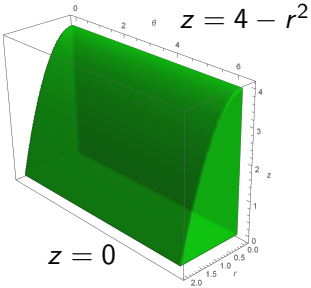
Use cylindrical coordinates to calculate

$$\iiint_D (x^2 + y^2)\,dV,$$

where D is the region defined by the inequalities

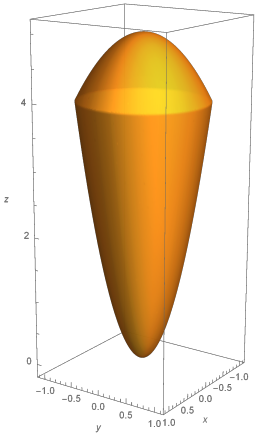
$$\begin{aligned} -2 \le x &\le 2, \\ -\sqrt{4 - x^2} \le y &\le \sqrt{4 - x^2}, \\ 0 \le z &\le 4 - x^2 - y^2. \end{aligned}$$





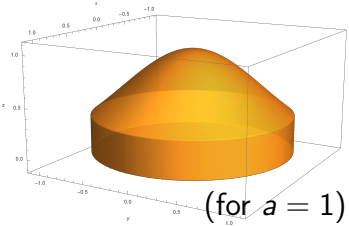
Example 12.12 (cf. Example 12.10)

Sketch and find the volume of the region between the two paraboloids $z = 4x^2 + 4y^2$ and $z = 5 - x^2 - y^2$ using cylindrical coordinates.



Example 12.13

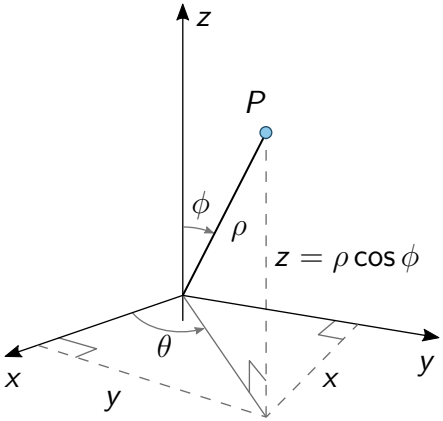
Find the volume of the region bounded by the cylinder $x^2 + y^2 = a$ $(a > 0)$ and the bell $0 \leq z \leq e^{-(x^2+y^2)}$.



(for $a = 1$)

Spherical coordinates

Spherical coordinates are useful for the integration over spherical domains or domains involving the intersections of spheres centred at the origin and circular cones with the vertex at the origin.

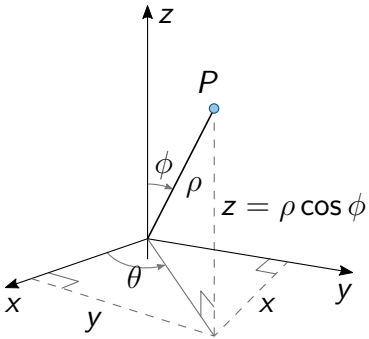


Change from Cartesian to spherical coordinates

The coordinate change from Cartesian (x, y, z) to spherical (ρ, θ, ϕ) is given by

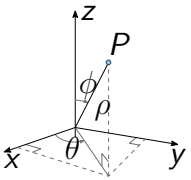
$$x = \rho \cos\theta \sin\phi,$$
$$y = \rho \sin\theta \sin\phi,$$
$$z = \rho \cos\phi.$$

Note that $\rho \geq 0$, $\theta \in [0, 2\pi]$, $\phi \in [0, \pi]$.



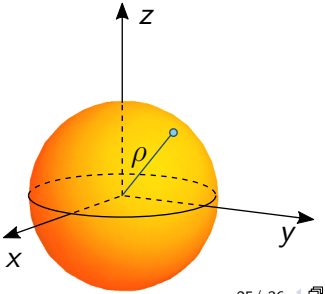
When ρ is constant

$$x = \rho\cos\theta\sin\phi,$$
$$y = \rho\sin\theta\sin\phi,$$
$$z = \rho\cos\phi.$$



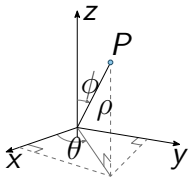
$$x^2 + y^2 + z^2 = \rho^2\sin^2\phi\cos^2\theta + \rho^2\sin^2\phi\sin^2\theta + \rho^2\cos^2\phi$$
$$= \rho^2\sin^2\phi + \rho^2\cos^2\phi = \rho^2.$$

The surfaces defined by constant ρ are the spheres of radius ρ centred at the origin. Hence, ρ can't be negative.

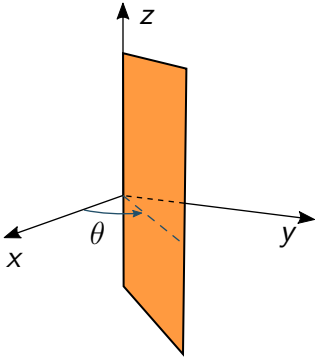


When θ is constant

$$x = \rho\cos\theta\sin\phi,$$
$$y = \rho\sin\theta\sin\phi,$$
$$z = \rho\cos\phi.$$

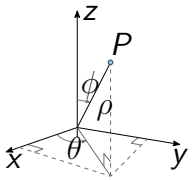


The coordinate θ is similar to θ in the polar and cylindrical coordinates, and ranges from 0 to 2π.
The surfaces defined by constant θ are the half-planes with boundary on the z-axis.



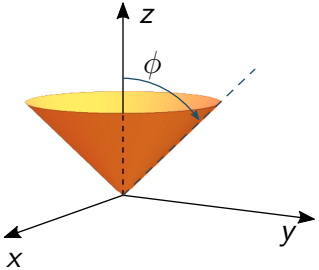
When ϕ is constant

$$x = \rho\cos\theta\sin\phi,$$
$$y = \rho\sin\theta\sin\phi,$$
$$z = \rho\cos\phi.$$



$$x^2 + y^2 = \rho^2\sin^2\phi = z^2\tan^2\phi,$$

The surfaces with constant ϕ are half-(circular) cones with the vertex in the origin and the axis coinciding with the z-axis. Hence the value of ϕ ranges from 0 to π.



Example 12.14

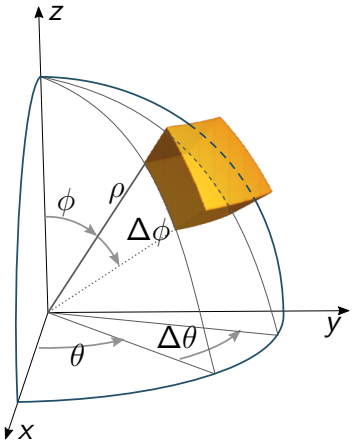
Find the equation of the sphere $x^2 + y^2 + (z-1)^2 = 1$ in spherical coordinates.

Example 12.15

Find a spherical coordinate equation for the cone $z = \sqrt{x^2 + y^2}$.

The Jacobian determinant for the spherical coordinates

$$\mathbf{F}\begin{pmatrix}\rho\\ \theta\\ \phi\end{pmatrix} = \begin{pmatrix}\rho\cos\theta\sin\phi\\ \rho\sin\theta\sin\phi\\ \rho\cos\phi\end{pmatrix},$$



$$|J_{\mathbf{F}}(\rho,\theta,\phi)| = \left|\det\begin{pmatrix}\cos\theta\sin\phi & -\rho\sin\theta\sin\phi & \rho\cos\theta\cos\phi\\ \sin\theta\sin\phi & \rho\cos\theta\sin\phi & \rho\sin\theta\cos\phi\\ \cos\phi & 0 & -\rho\sin\phi\end{pmatrix}\right| = |\rho^2\sin\phi| = \rho^2\sin\phi.$$

Example 12.16

Using spherical coordinates, evaluate the volume of a ball of radius 1.

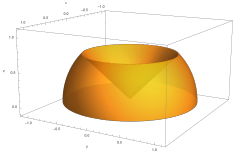
Example 12.17

Let B be the unit ball defined by the equation $x^2 + y^2 + z^2 \leq 1$. Evaluate the integral

$$\iiint_B e^{(x^2+y^2+z^2)^{\frac{3}{2}}}\, dV.$$

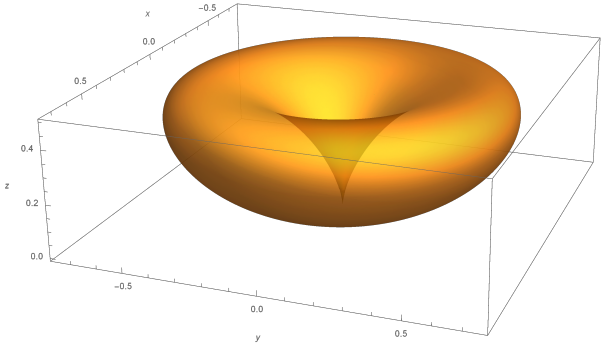
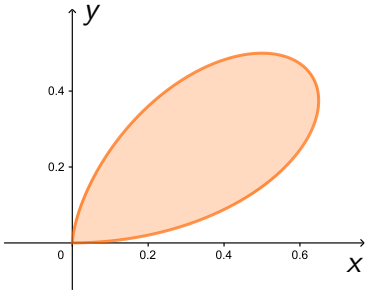
Example 12.18

Find the volume of the solid enclosed by the hemisphere $z = \sqrt{4 - x^2 - y^2}$, bounded from above by the cone $z = \sqrt{x^2 + y^2}$ and below by the xy-plane.



Example 12.19

Find the volume of the solid D enclosed by the surface

$$(x^2 + y^2 + z^2)^2 = 2z(x^2 + y^2)$$

using spherical coordinates.